

Wedding Photo Sharing & Face Search Application – Product Requirements Document (PRD)

1. Overview

A wedding photo sharing platform where guests can:

1. View and download wedding photos.
2. Upload a selfie.
3. Automatically find photos in which they appear.
4. Share albums with family and friends.
5. Access photos without manually browsing thousands of images.

The platform should support fetching photos from:

- Google Drive
 - Google Photos
 - Any web server/CDN
 - Optional future support for cloud storage such as Amazon S3 and Microsoft Azure Blob Storage
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2. User Personas

Wedding Guest

- Upload selfie
- View matching photos
- Download photos
- Create favorites list
- Share photos

Bride/Groom

- Create event
- Manage albums
- Upload photos

- Control access
- View analytics

Photographer

- Upload photos
- Sync photos from storage
- Manage albums

Administrator

- Manage events
 - Monitor processing
 - Manage storage and subscriptions
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3. Key Features

Feature 1: Wedding Event Management

Create Event

Fields:

- Event Name
- Bride Name
- Groom Name
- Event Date
- Cover Photo
- Access Code
- Event URL

Example:

Rahul & Priya Wedding

wedding.app/rahul-priya

Feature 2: Photo Source Integration

Source Types

Google Drive Integration

Admin provides:

- Folder ID
- OAuth credentials

System:

- Reads images
- Syncs metadata
- Downloads thumbnails
- Processes faces

Google Photos Integration

Admin connects Google Photos account.

System:

- Reads selected albums
- Syncs photos
- Processes new photos automatically

Web Server Integration

Example:

<https://photos.mydomain.com/wedding/>

System:

- Scan directory
 - Import photos
 - Generate thumbnails
 - Index faces
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Feature 3: Face Recognition Search

Guest Journey

1. Open wedding website.
2. Upload selfie.
3. Face is extracted.
4. Similar faces are searched.
5. Matching photos are shown.

Example:

Upload Selfie

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Face Encoding

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Vector Search

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Top Matching Photos

Results

Display:

- Match confidence
- Photo thumbnail
- Download option
- Share option

Feature 4: AI Face Processing Pipeline

During Import

For every photo:

Photo

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Face Detection

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Face Embedding

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Store Vector

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Search Ready

Store:

- Face vector
- Face bounding box
- Event ID
- Photo ID

Technologies

Face Detection:

- RetinaFace
- MTCNN
- InsightFace

Face Embedding:

- ArcFace
- FaceNet
- InsightFace

Vector Search:

- FAISS
- Qdrant
- Weaviate

Recommended:

- InsightFace + Qdrant

Feature 5: Guest Access

Option A

Access Code

Enter Wedding Code

Option B

Magic Link

Email

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OTP

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Access

Option C

Public Album

No login required.

Feature 6: Album Browsing

Filters:

- Ceremony
- Sangeet
- Mehendi
- Haldi
- Reception
- Family Photos

Sorting:

- Latest
- Popular
- Photographer Choice

Feature 7: Favorites

Guest can:

 Favorite photos

Later:

My Favorites

Feature 8: Download & Sharing

Single Download

Download original image.

Bulk Download

Download all matched photos as ZIP.

Share

Generate shareable link.

Feature 9: Photographer Dashboard

Upload

- Drag & drop
- Folder upload
- Sync from Google Drive
- Sync from Google Photos

Progress

2000 Photos Uploaded

1780 Processed

220 Processing

Feature 10: QR Code Access

Wedding invitation includes QR code.

Guest scans:

QR

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Wedding Album

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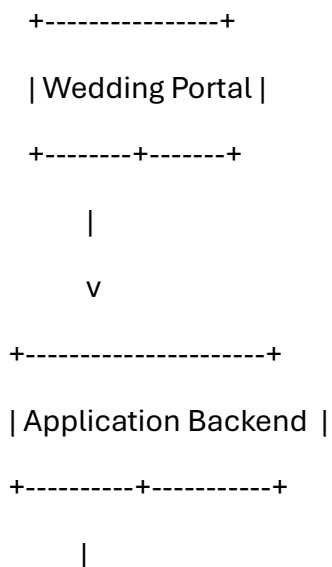
Upload Selfie

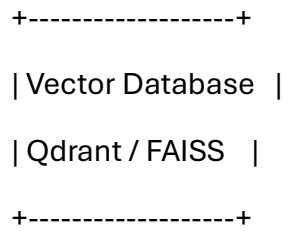
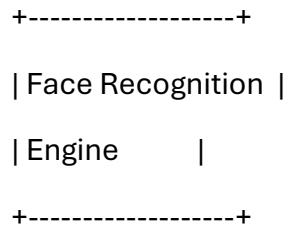
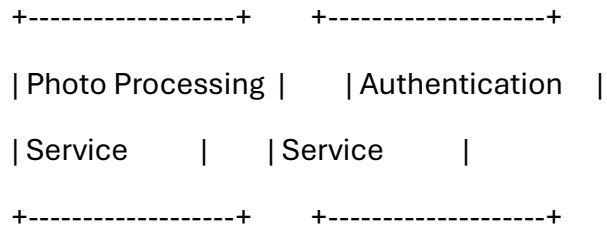
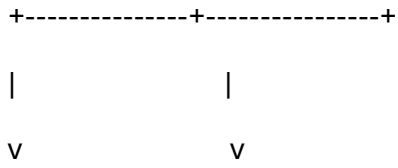
↓

See Your Photos

4. Architecture

High Level





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| Photo Storage |

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5. Recommended Tech Stack

Frontend

- Next.js
- React
- Tailwind CSS

Backend

- Node.js
- NestJS or Express

Face Recognition

- InsightFace

Vector Database

- Qdrant

Database

- PostgreSQL

Storage

- Google Drive
 - Google Photos
 - Amazon S3
 - Azure Blob Storage
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6. Database Model

Events

id

name

event_date

slug

cover_image

access_code

Photos

id

event_id

source_url

thumbnail_url

capture_time

Faces

id

photo_id

vector_id

face_box

Users

id

email

name

Searches

id

user_id

uploaded_selfie

search_time

7. Non-Functional Requirements

Scale

- 50,000+ photos per wedding
- 10,000 guests
- 500 concurrent users

Performance

- Selfie search < 5 seconds
- Thumbnail load < 1 second
- Face indexing < 2 seconds/photo

Availability

- 99.9% uptime

8. Privacy & Security

Consent

Wedding owner confirms photo-sharing consent.

Face Data

- Encrypt embeddings at rest
- Auto-delete uploaded selfies after search
- Allow guest to request removal

Access Controls

- Password-protected events
- OTP access
- Private albums

9. Future Enhancements

AI Features

- Group photo detection
- Family relationship tagging
- Auto-generated highlights
- Best smile selection
- Duplicate detection

Wedding Assistant

Ask:

"Show all photos from Sangeet where I appear with the bride."

Using multimodal search over photos and facial recognition.

MVP (6–8 Weeks)

1. Event creation
2. Google Drive sync
3. Web server sync
4. Selfie upload
5. Face recognition search
6. Gallery view
7. Download photos
8. QR code access

This MVP is sufficient to compete with services like facial-search wedding galleries while keeping infrastructure costs relatively low.